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Experiment No. 9

Familiarization with the Alternating Current (AC) Waves Using Software (LTSpice) Simulation.

Objective

In this experiment, we shall study some aspects of a sinusoidal waveform, and correlate these with practically measurable values such as peak value, phase angle, and time period. We shall apply a sinusoidal wave input to an RC circuit and observe the effect of changing the frequency on the phase angles.

Theory

Any periodic variation of current or voltage where the current (or voltage), when measured along any particular direction goes positive as well as negative, is defined to be an AC quantity. Sinusoidal AC wave shapes are the ones where the variation (current or voltage) is a sine function of time.

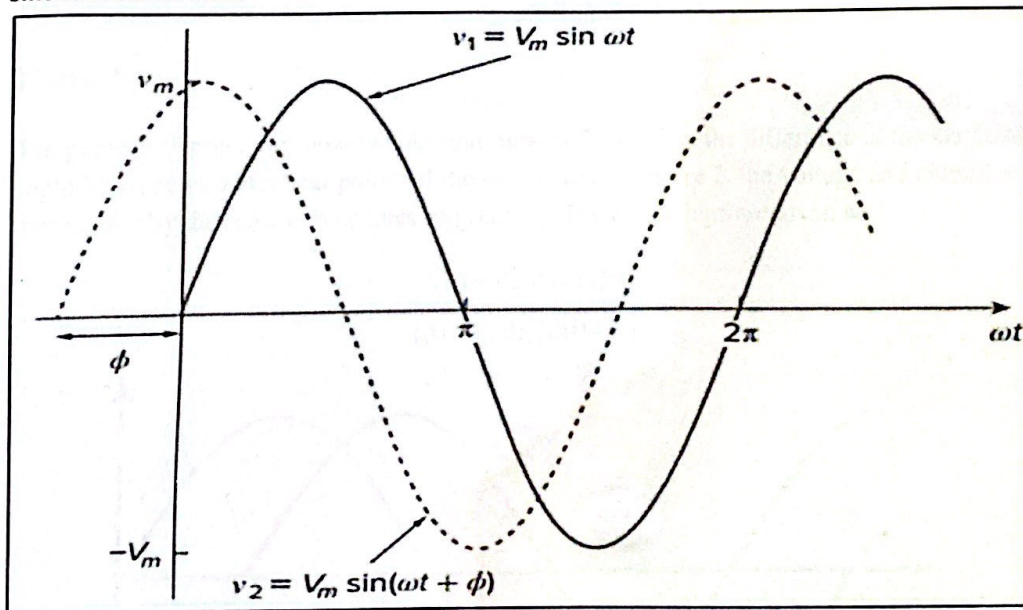


Figure 1: A Sine wave

Here, the time period = T

Frequency, $f = \frac{1}{T}$

$$V_1(t) = V_m \sin(2\pi ft) = V_m \sin(\omega t)$$

$$\text{and, } V_2(t) = V_m \sin(2\pi ft + \phi) = V_m \sin(\omega t + \phi)$$

Effective Value

The general equation of the RMS value of any function (voltage, current, or any other physical quantity for which rms. calculation is meaningful) is given by the equation,

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T V^2 dt}$$

Now, for sinusoidal functions, using the above equation we get the RMS value by dividing the peak value (V_m) by the square root of 2. That is,

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T (V_m \sin(2\pi ft))^2 dt} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (V_m \sin \theta)^2 d\theta} = \frac{V_m}{\sqrt{2}}$$

Similarly, for currents, $I = I_m/\sqrt{2}$. These RMS values can be used directly for power

calculation. The formula for average power is given by $P_{avg} = \sqrt{\frac{1}{T} \int_0^T (VI) dt}$. And for sinusoids this leads to $P_{avg} = VI \cos \theta$. Here, V and I are rms values and θ is the phase angle between voltage wave and current wave. The phase angle is explained in the next section.

Phase Angle

The phase difference between two ac sinusoidal waveforms is the difference in the electrical angle between two identical points of the two waves. In figure 2, the voltage and current are represented by the red and blue lines respectively. Their equations are given as:

$$V(t) = V_m \sin(2\pi ft)$$

$$I(t) = I_m \sin(2\pi ft - \theta)$$

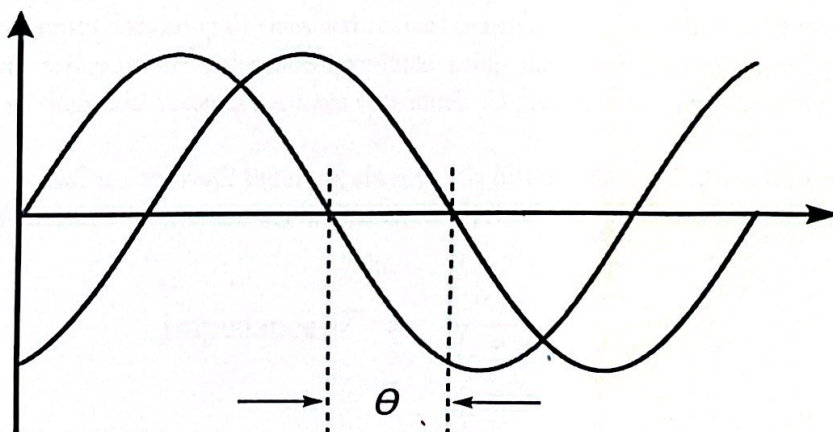


Figure 2: Phase Angle between two signals

It is worth noting that one of the signals would be leading the other one. In other words, the other signal would be lagging the first one. The phase difference can be determined by observing the difference between the time/phase corresponding to the peaks (either both positive or both negative) of the signals, or, alternatively, the zero crossing times from the same side.

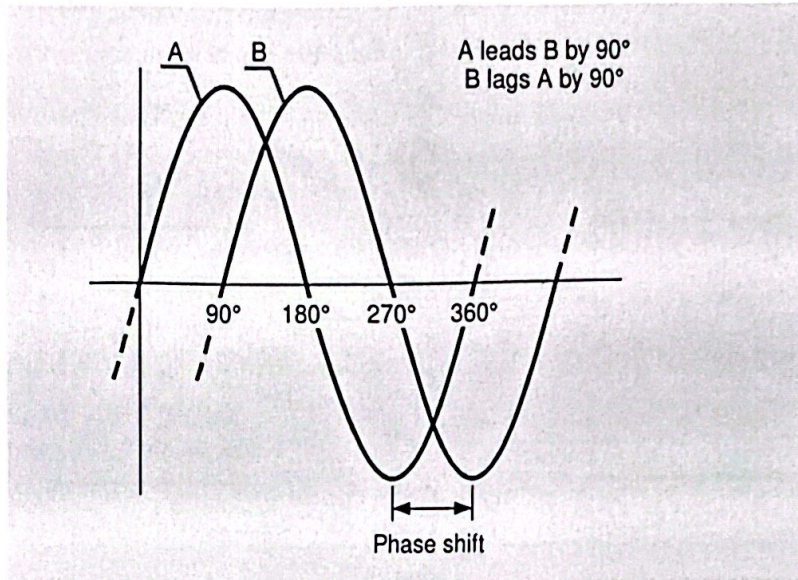


Figure 3: Concept of Leading & Lagging

If the time difference between the two positive peaks/negative peaks/zero crossings is found to be t_{diff} , Then the **Phase Difference** = $t_{diff} * f * 360^\circ$, where, f = frequency.

Impedance

For AC circuit analysis, impedance plays the same role as resistance plays in DC circuit analysis. It can be stated safely, that the concept of impedance is the most important thing that makes AC analysis so much popular among engineers. As you will see in your later courses, any other periodic forms of time-varying voltages or currents, are converted into an equivalent series consisting of sines and cosines (much like any function can be expanded by the power series of the independent variable using the Taylor series), only because the analysis of sinusoidal voltages becomes very much simple due to the impedance technique.

What is impedance anyway? Putting it simply, it is just the ratio of RMS voltage across the device to the RMS current through it. That is:

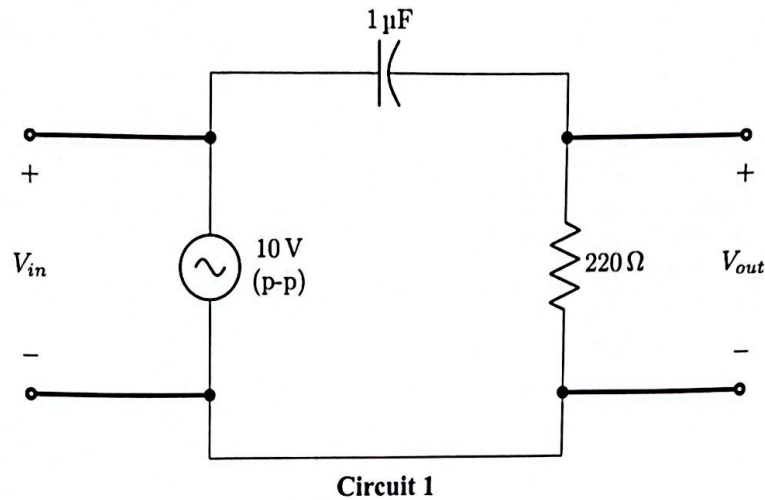
$$\text{Impedance, } Z = \frac{V}{I \angle P} = \frac{V_m}{I_m \angle P}$$

Its unit is Ohms (Ω).

Procedure

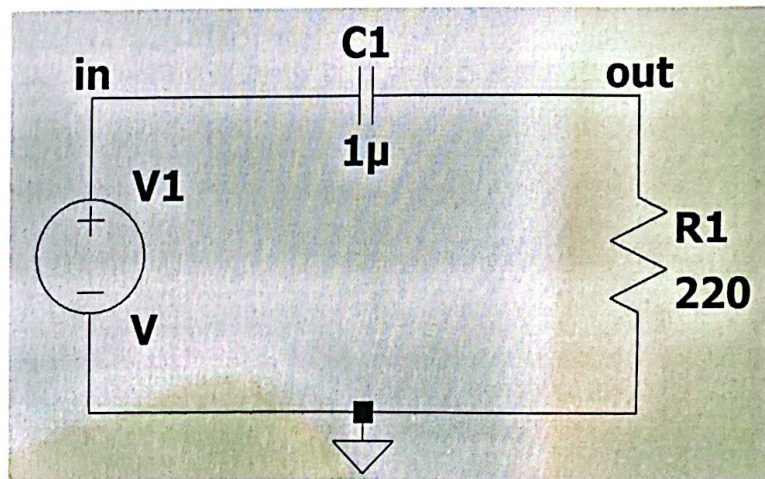
Simulation using LTSpice

We will familiarize you with the Alternating Current (AC) Waves in LTSpice by simulating the simple circuit shown below (**Circuit 1**). Let us visualize V_{in} and V_{out} and find out the phase difference introduced by the capacitor in this circuit..



Let us simulate this circuit step by step in LTSpice as described below:

- Open a new schematic window by clicking **File** → **New Schematic**. Draw the circuit shown below in **Figure 1**. Modify the components with their values and name the nodes. To name the nodes, **Right-click on the wire/node** → **Label Net**. Do not forget to add a ground to the circuit.



- To modify the voltage source as an AC voltage source, **Right-click on the voltage source** → **Select Advanced** → **insert the values as below and, click OK-** for a 10V p-p 1 KHz AC voltage source.

Independent Voltage Source - V1

Functions

☐ (none)

☐ PULSE(V1 V2 Tdelay Trise Tfall Ton Period Ncycles)

☒ SINE(Voffset Vamp Freq Td Theta Phi Ncycles)

☐ EXP(V1 V2 Td1 Tau1 Td2 Tau2)

☐ SFFM(Voff Vamp Fcar MDI Fsig)

☐ PWL(t1 v1 t2 v2...)

☐ PWL FILE:

DC offset[V]:

Amplitude[V]:

Freq[Hz]:

Tdelay[s]:

Theta[1/s]:

Phi[deg]:

Ncycles:

Additional PWL Points

Make this information visible on schematic: ☒

DC Value

DC value:

Make this information visible on schematic: ☒

Small signal AC analysis(AC)

AC Amplitude:

AC Phase:

Make this information visible on schematic: ☒

Parasitic Properties

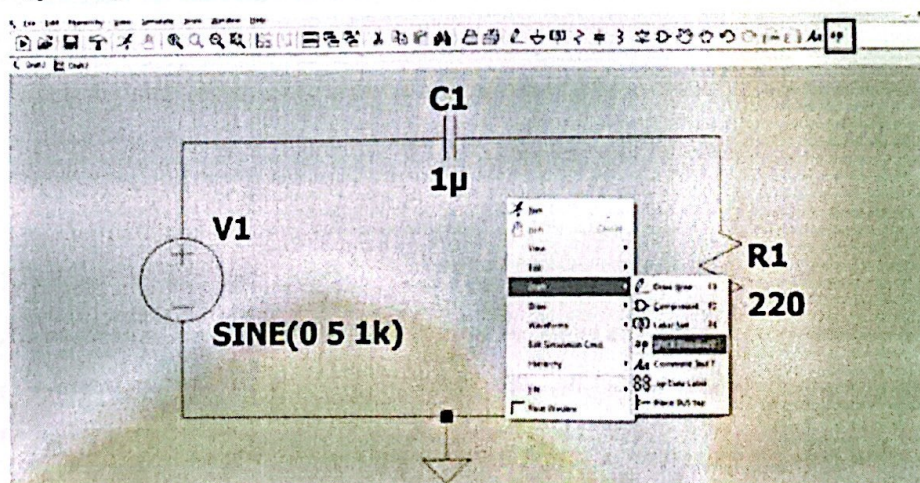
Series Resistance[Ω]:

Parallel Capacitance[F]:

Make this information visible on schematic: ☒

- To see the responses/waveshapes we must do '*Transient Analysis*'. The transient analysis calculates a circuit's response over a period.

To run the transient analysis, we have to write the analysis command. Find the ***'Spice Directives'*** option by *Right-clicking on the schematic → Draft → Spice Directives* or, *find it from the toolbar above.*



- After clicking the '*Spice Directives*', the '*Edit Text on the Schematic*' window will appear. Now **Right-click on the blank space on this window** → **Select 'Help me Edit'** → **Analysis Command**. A window titled '*Edit Simulation Command*' will appear. Insert values in the boxes as below and click OK. It will generate a transient analysis command. Place the command somewhere on the schematic. *[Notice the '.tran' syntax for transient analysis.]*

Edit Simulation Command

Transient AC Analysis DC sweep Noise DC Transfer DC op prt

Perform a non-linear, time-domain simulation.

Stop time: 8ms

Time to start saving data:

Maximum Timestep: 1us

Start external DC supply voltages at 0V: ☐

Stop simulating if steady state is detected: ☐

Don't reset T=0 when steady state is detected: ☐

Step the load current source: ☐

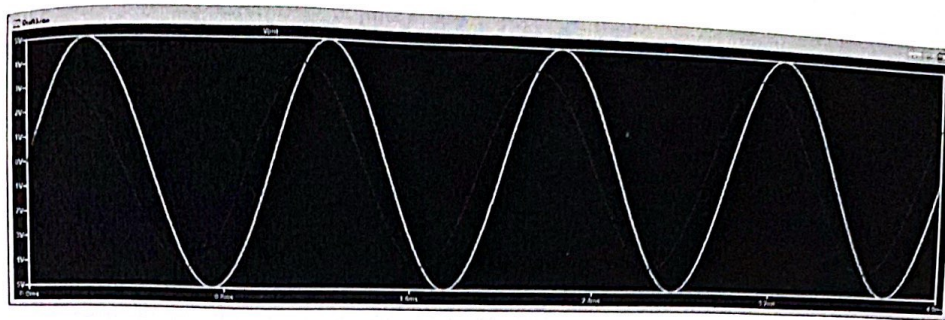
Skip initial operating point solution: ☐

Syntax: .tran <Tprint> <Tstop> [<Tstart> [<Tmaxstep>]] [<option> [<option>] ...]

.tran 0 8ms 0 1us

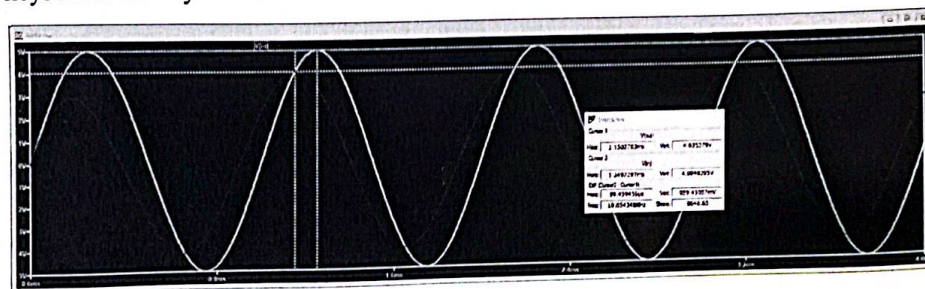
Cancel OK

- To run the simulation, click '*Run*'. Find the '*Run*' button from the above toolbar or by just Right-clicking on the schematic.
- After clicking the '*Run*' button a plot window will appear. In this window we can see responses and waveshapes of voltage and currents with respect to time. To see a plot, **Right-click on the plot window** → **Add trace** → **Select any voltage or current** → **OK**. *[We can also add trace by simply using markers on the schematic. When the run is complete a cursor will appear if we place the mouse cursor on a wire or component of the circuit.]*



The axes properties (Range) can be changed by **Right-clicking on the horizontal (x-axis) and the vertical (y-axis).**

- To extract data from a plot/response, use the data cursor. A cursor for a particular trace will appear by clicking on the name of that trace. One click will produce one cursor, clicking twice will produce two. Two produce two cursors for two different traces, first, click twice on the name of one trace. Then, click once on the name of the other trace. The data point of the cursor can be moved by the arrow keys from the keyboard.



- A window will appear on the bottom right corner containing the values corresponding to the cursors. Note that it also shows the difference between the two cursors (data points) for both the vertical and the horizontal axes. Use this to find **Time difference**, t_{diff} between V_{in} & V_{out} (Notice that V_{out} is the representative of supply current in this circuit in mA). Use t_{diff} to determine **Phase Difference**.
- Save the Schematic by clicking **File** → **Save as** → '**Name.asc**' and the plots by clicking **File** → **Save plot settings** → '**Name.plt**' for future use and analysis.

Lab Work

1. Measure the Phase Difference between V_{in} & V_{out} from the plot in degrees.

From the Circuit we simulated, the Phase Difference =

35.92°

2. Perform similar analysis to Measure the Phase Difference between V_{in} & V_{out} when the frequency of the supply voltage is 500 Hz. Do the same for 2 KHz. Observe the effect of changing the frequency on the Phase Difference.
3. State your opinion on your observations on the effect of changing the frequency on the Phase Difference. Does it make any difference?

if frequency increases the phase difference decreases

for 1kHz we get 35.92°

for 2kHz we get $2000 \times 360 \times 25.14 \times 10^{-6} = 18.1^\circ$

for 0.5kHz " " $500 \times 360 \times 308.6 \times 10^{-6} = 55.5^\circ$

for 1kHz " " $1000 \times 360 \times 99.76 \times 10^{-6} = 35.92^\circ$

Report

1. Answer to questions and Complete the Lab work sections.
2. Save all your .asc and .plt files and make a zip file. You need to submit it with the report.
3. Discussion [comment on the obtained results and discrepancies]. Add pages if necessary.

here to get the phase difference we used the formula $[PD = 4\pi f \times 360^\circ]$ by which we get the values in different frequencies.

